1.32 Differentiation: Tangents and Normals

Worked Solutions

1a)

$$f'(x) = 3x^2 + 2x^{-2}, \qquad f'(2) = 12 + \frac{1}{2} = \frac{25}{2}.$$

1b)

$$f'(x) = \frac{1}{2}x^{-1/2} + 8x^3, \qquad f'(2) = \frac{1}{2\sqrt{2}} + 64 = \frac{\sqrt{2}}{4} + 64.$$

1c)

$$f(x) = x^3 - x^{-2}, \quad f'(x) = 3x^2 + 2x^{-3}, \quad f'(2) = 12 + \frac{1}{4} = \frac{49}{4}.$$

1d)

$$f(x) = x^{1/2} - x^{3/2}, \quad f'(x) = \frac{1}{2}x^{-1/2} - \frac{3}{2}x^{1/2}.$$

$$f'(2) = \frac{1}{2\sqrt{2}} - \frac{3}{2}\sqrt{2} = -\frac{5\sqrt{2}}{4}.$$

2a)

$$\frac{dy}{dx} = 2x, \qquad \left.\frac{dy}{dx}\right|_{x=2} = 4.$$

$$y - 4 = 4(x - 2) \Rightarrow y - 4 = 4x - 8 \Rightarrow 4x - y - 4 = 0.$$

2b)

$$\frac{dy}{dx} = 3x^2 - 1, \qquad \left.\frac{dy}{dx}\right|_{x=1} = 2.$$

$$y - 5 = 2(x - 1) \Rightarrow y - 5 = 2x - 2 \Rightarrow 2x - y + 3 = 0.$$

2c)

$$y = x^{-1} \Rightarrow \frac{dy}{dx} = -x^{-2}, \qquad \left.\frac{dy}{dx}\right|_{x=-2} = -\frac{1}{4}.$$

$$y + \frac{1}{2} = -\frac{1}{4}(x + 2) \Rightarrow 4y + 2 = -x - 2 \Rightarrow x + 4y + 4 = 0.$$

2d)

When $x = 1$, $y = 2$.

$$y = 2x^{1/2} \Rightarrow \frac{dy}{dx} = x^{-1/2}, \qquad \left.\frac{dy}{dx}\right|_{x=1} = 1.$$
$$y - 2 = 1(x - 1) \Rightarrow y - 2 = x - 1 \Rightarrow x - y + 1 = 0.$$

3a)

When $x = 0$, $y = 3$.

$$\frac{dy}{dx} = -2x, \qquad \left.\frac{dy}{dx}\right|_{x=0} = 0.$$
$$y - 3 = 0(x - 0) \Rightarrow y - 3 = 0.$$

3b)

The gradient is zero, so the curve is neither increasing nor decreasing at $x = 0$ (it is the maximum point of the quadratic).

4a)

$$\frac{dy}{dx} = 6x, \qquad \left.\frac{dy}{dx}\right|_{x=1} = 6.$$
$$y - 3 = -\frac{1}{6}(x - 1) \Rightarrow 6y - 18 = -x + 1 \Rightarrow x + 6y - 19 = 0.$$

4b)

$$\frac{dy}{dx} = 20x^3 - 2x, \qquad \left.\frac{dy}{dx}\right|_{x=-1} = -18.$$
$$y - 4 = \frac{1}{18}(x + 1) \Rightarrow 18y - 72 = x + 1 \Rightarrow x - 18y - 73 = 0.$$

4c)

$$y = x^{-2} \Rightarrow \frac{dy}{dx} = -2x^{-3}, \qquad \left.\frac{dy}{dx}\right|_{x=2} = -\frac{1}{4}.$$
$$y - \frac{1}{4} = 4(x - 2) \Rightarrow y - \frac{1}{4} = 4x - 8 \Rightarrow 4y - 1 = 16x - 32 \Rightarrow 16x - 4y - 31 = 0.$$

4d)

$$y = x^{1/2} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^{-1/2}, \qquad \left.\frac{dy}{dx}\right|_{x=1} = \frac{1}{2}.$$
$$y - 1 = -2(x - 1) \Rightarrow y - 1 = -2x + 2 \Rightarrow 2x + y - 3 = 0.$$

4e)

When $x = -1$, $y = -1 - 1 = -2$.

$$\frac{dy}{dx} = 1 - 4x^3, \qquad \left.\frac{dy}{dx}\right|_{x=-1} = 5.$$
$$y + 2 = -\frac{1}{5}(x + 1) \Rightarrow 5y + 10 = -x - 1 \Rightarrow x + 5y + 11 = 0.$$

5)

$$\frac{dy}{dx} = 10x = 30 \Rightarrow x = 3.$$

6)

$$\frac{dy}{dx} = 3x^2 - 2 = 10 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2.$$

7)

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{4\sqrt{2}} \Rightarrow \sqrt{x} = 2\sqrt{2} \Rightarrow x = 8.$$

8a)

$$y = x^{-1} + 2, \quad \frac{dy}{dx} = -x^{-2}, \quad \left.\frac{dy}{dx}\right|_{x=1} = -1.$$
$$y - 3 = -1(x - 1) \Rightarrow y - 3 = -x + 1 \Rightarrow y = -x + 4.$$

8b)

$$y - 3 = 1(x - 1) \Rightarrow y - 3 = x - 1 \Rightarrow y = x + 2.$$

8c)

Tangent line meets axes at $(0, 4)$ and $(4, 0)$.

Normal line meets axes at $(0, -2)$ and $(-2, 0)$.

9a)

$$\frac{dy}{dx} = 3x^2 - 6x, \qquad \left.\frac{dy}{dx}\right|_{x=3} = 9.$$
$$y - 2 = -\frac{1}{9}(x - 3) \Rightarrow 9y - 18 = -x + 3 \Rightarrow x + 9y = 21.$$

9b)

$$\left(0, \frac{7}{3}\right) \text{ and } (21, 0).$$

9c)

$$\frac{1}{2} \times \frac{7}{3} \times 21 = \frac{49}{2} = 24.5 \text{ units}^2.$$

10a)

$$y = x^{-1} + x^3, \quad \frac{dy}{dx} = -x^{-2} + 3x^2, \quad \left.\frac{dy}{dx}\right|_{x=1} = 2.$$
$$y - 3 = 2(x-1) \Rightarrow y - 3 = 2x - 2 \Rightarrow y = 2x + 1.$$

10b)

When $x = 0$, $y = 0$ so $(0, 1)$.

When $y = 0$, $x = -\frac{1}{2}$ so $\left(-\frac{1}{2}, 0\right)$.

10c)

$$2x + 1 = \frac{1}{x} + x^3 + 1 \Rightarrow 2x^2 + x = 1 + x^4 + x.$$
$$\Rightarrow x^4 - 2x^2 - x + 1 = 0 \Rightarrow (x^2 - 1)(x^2 - 1) = 0.$$
$$\Rightarrow (x-1)^2(x+1)^2 = 0 \text{ so } x = \pm 1.$$

When $x = -1$, $y = -1 - 1 = -1$ so $(-1, -1)$.

11a)

$$y = 5x^2 - x^{-1}, \qquad \frac{dy}{dx} = 10x + x^{-2}.$$

$y + 9x = 5 \Rightarrow y = -9x + 5$ so the gradient is -9.

$$10x + x^{-2} = -9 \Rightarrow 10x^3 + 1 = -9x^2 \Rightarrow 10x^3 + 9x^2 + 1 = 0.$$
$$\Rightarrow (x+1)(10x^2 - x + 1) = 0 \text{ has solution } x = -1 \text{ so A is } (-1, 6).$$

11b)

$$y - 6 = -9(x+1) \Rightarrow y = -9x - 3.$$

11c)

$$5x^2 - x^{-1} = -9x - 3 \Rightarrow 5x^3 - 1 = -9x^2 - 3x.$$
$$\Rightarrow 5x^3 + 9x^2 + 3x - 1 = 0 \Rightarrow (x+1)^2(5x-1) = 0.$$

So when $x = \frac{1}{5}$, i.e. $\left(\frac{1}{5}, -\frac{24}{5}\right)$.

12)

$$y = kx^{-1} \Rightarrow \frac{dy}{dx} = -kx^{-2}, \qquad \left.\frac{dy}{dx}\right|_{x=1/2} = -4k.$$

13)

$$1 = 8 + 4p + 2q + 5 \Rightarrow 4p + 2q = -12.$$
$$\frac{dy}{dx} = 3x^2 + 2px + q.$$
$$10 = 12 + 4p + q \Rightarrow 4p + q = -2.$$
$$p = 2 \text{ and } q = -10.$$

14)

At $y = 0$, $0 = -x^2 + 2x + 8$, so $x = -2$ and $x = 4$.

A$(-2, 0)$ and B$(4, 0)$.

$$f'(x) = -2x + 2, \qquad f'(-2) = 6.$$
$$y - 0 = 6(x + 2) \Rightarrow y = 6x + 12.$$
$$f'(4) = -6, \quad y - 0 = -6(x - 4) \Rightarrow y = -6x + 24.$$
$$6x + 12 = -6x + 24 \text{ so } 12x = 12 \Rightarrow x = 1.$$
$$y = 6 \times 1 + 12 = 18.$$

Area of triangle: $\frac{1}{2} \times 6 \times 18 = 54$ units^2.

15)

$$6 = -1 + (1 + p) - q + 5 \Rightarrow p - q = 1.$$
$$\frac{dy}{dx} = 3x^2 + 2(1 + p)x + q.$$
$$-2 = 3 - 2(1 + p) + q \Rightarrow -2p + q = -3.$$
$$p = 2 \text{ and } q = 1.$$

16)

$$y = -3 + 5 + 1 = 3.$$
$$y = 3x^3 - 5x^5 + 1 \Rightarrow \frac{dy}{dx} = 9x^2 - 25x^4.$$

Gradient at $(-1, 3)$: $\left.\frac{dy}{dx}\right|_{x=-1} = -16$.

$$9x^2 - 25x^4 = -16 \Rightarrow 25x^4 - 9x^2 - 16 = 0.$$
$$(x - 1)(x + 1)(25x^2 + 16) = 0.$$

$x = 1$ and $x = -1$ so at $x = 1$, $y = 3 - 5 + 1 = -1$.

$$y + 1 = -16(x - 1) \Rightarrow y = -16x + 15.$$

17)

$$y = x^{-2} \Rightarrow \frac{dy}{dx} = -2x^{-3}.$$

When $x = a$, $y = \frac{1}{a^2}$.

$$y - \frac{1}{a^2} = -\frac{2}{a^3}(x - a).$$
$$a^3y - a = -2(x - a), \quad a^3y - a = -2x + 2a, \quad a^3y + 2x = 3a.$$

18a)

$$y = x^3 + 2x \Rightarrow \frac{dy}{dx} = 3x^2 + 2.$$

When $x = a$, $y = 3a^2 + 2$.

$$y - (a^3 + 2a) = (3a^2 + 2)(x - a).$$
$$y - a^3 - 2a = 3a^2x - 3a^3 + 2x - 2a.$$
$$y = (3a^2 + 2)x - 2a^3.$$

18b)

$x^2 \geq 0$ for all real values of x, so $3x^2 + 2 > 0$ for all real values of x, so the gradient of the curve is always positive.

19)

$$y = x^2 - kx^3 \Rightarrow \frac{dy}{dx} = 2x - 3kx^2.$$
$$\Rightarrow \frac{d^2y}{dx^2} = 2 - 6kx \text{ so } \left.\frac{d^2y}{dx^2}\right|_{x=2} = 2 - 12k.$$

20a)

$$f'(x) = 6e^{6x} \text{ so } f'(1) = 6e^6.$$

20b)

$$f'(x) = \frac{1}{2}x^{-1/2} - 2e^{2x} \text{ so } f'(1) = \frac{1}{2} - 2e^2.$$

20c)

$$f'(x) = \frac{4}{3}x^{1/3} + 2e^{2x} \text{ so } f'(1) = \frac{4}{3} + 2e^2.$$

21)

$$\frac{dy}{dx} = e^x, \qquad \left.\frac{dy}{dx}\right|_{x=0} = 1.$$
$$y - 1 = 1(x - 0) \text{ so } y = x + 1.$$

22)

$$\frac{dy}{dx} = 2x - 2e^{2x}, \qquad \left.\frac{dy}{dx}\right|_{x=0} = -2.$$
$$y + 1 = \frac{1}{2}(x - 0) \text{ so } y = \frac{1}{2}x - 1.$$

23a)

$$\frac{dy}{dx} = e^x - 4x, \qquad \left.\frac{dy}{dx}\right|_{x=1} = e - 4.$$

$$y - (e-2) = (e-4)(x-1).$$

$$\Rightarrow y = (e-4)x + (4-e) + e - 2 \Rightarrow y = (e-4)x + 2.$$

23b)

$$(0,2) \text{ and } \left(-\frac{2}{e-4}, 0\right) = \left(\frac{2}{4-e}, 0\right).$$

23c)

$$\frac{1}{2} \times 2 \times \frac{2}{4-e} = \frac{2}{4-e}.$$

24a)

$$\frac{dy}{dx} = -4ke^{-kx}, \qquad \left.\frac{dy}{dx}\right|_{x=2} = -4ke^{-2k}.$$

24b)

$$\frac{d^2y}{dx^2} = 4k^2e^{-kx} = k^2y \text{ as req.}$$

25)

$$y = e^{5kx} - ke^{5kx}, \quad \frac{dy}{dx} = 5ke^{5kx} - 5k^2e^{5kx}.$$

$$\frac{d^2y}{dx^2} = 25k^2e^{5kx} - 25k^3e^{5kx} = 25k^2e^{5kx}(1-k).$$

When $x = \frac{1}{k}$, $25k^2e^{5k\times 1/k}(1-k) = -100e^5$.

$$25k^2(1-k) = -100 \Rightarrow k^2 - k^3 = -4 \Rightarrow k^3 - k^2 - 4 = 0.$$

$$(k-2)(k^2+k+2) = 0 \text{ so } k = 2.$$

26a)

$$\frac{dy}{dx} = 5 - e^x, \qquad \left.\frac{dy}{dx}\right|_{x=2} = 5 - e^2.$$

$$y - (10 - e^2) = \frac{1}{e^2-5}(x-2).$$

$$\Rightarrow y = \frac{1}{e^2-5}x - \frac{2}{e^2-5} + 10 - e^2.$$

26b)

$$\left(0, -\frac{2}{e^2-5} + 10 - e^2\right).$$

$$\frac{2}{e^2-5} - 10 + e^2 = \frac{1}{e^2-5}x.$$

$$\Rightarrow x = 2 - 10(e^2-5) + e^2(e^2-5) \Rightarrow x = e^4 - 15e^2 + 52.$$

So $(e^4 - 15e^2 + 52, 0)$.

27)

$$f'(x) = e^x - 4x - 1 - e^{-x} \text{ so } f'(0) = -1.$$

At $x = 0$, $y = 2$ so $y - 2 = 1(x - 0) \Rightarrow y = x + 2$.

$$g(x) = \frac{1}{2}x^2 + (5+k)x + p, \quad g'(x) = x + 5 + k.$$

$$g'(7) = 1 = 12 + k \text{ so } k = -11 \Rightarrow g(x) = \frac{1}{2}x^2 - 6x + p.$$

At $(7, 9)$, $9 = \frac{49}{2} - 42 + p$ so $p = 26.5$.